Group 3: Exploring Pricing Strategies for PizzaHub

Objective:

Your team will assist **PizzaHub**, a popular pizza store that offers delivery services, in optimizing its pricing strategies. PizzaHub sells a single type of pizza and delivers to customers across four regions. On **March 1st, 2023**, PizzaHub introduced a **5% discount on food fees** in **Region 3,** and a **5% discount on delivery fees in Region 4** to encourage more orders in these areas. **Regions 1 and 2** did not receive any discounts.

PizzaHub wants to refine its pricing strategies based on the historical data to maximize profits in Region 3 and Region 4.

Using the provided dataset, GP3Data.csv, complete the following tasks:

1. **Estimate the Demand Functions**: Analyze the data to determine the demand functions.
2. **Optimize Profitability**: Based on your demand analysis, recommend an optimized pricing strategy that maximizes PizzaHub's profit across Region 3 and Region 4.

Data and Variables:

The dataset contains daily information on each order received by PizzaHub from **January 2023 to April 2023**. Key variables include:

1. **Date of the order**
2. **Region**: The region where the order was placed.
3. **Unit Price**: The price of a single pizza.
4. **Quantity ordered**
5. **Distance**: The distance from the store to the delivery destination.
6. **Delivery Fees:** Calculated as $0.5 per mile based on the distance.
7. **Discount Applied**: Indicates if the 5% discount was applied to the order.

Production Costs:

PizzaHub incurs two types of costs:

1. **Fixed Costs**: Includes equipment, rent, and salaries, totaling **$5,000 per year**.
2. **Variable Costs**:
    - **Pizza Production Cost**: $4 per pizza.
    - **Delivery Cost**: $0.5 per mile for each delivery.

General Guidelines:

- **Report Requirements**: Keep your report concise and professional, within four pages (excluding appendices).
- **Collaboration**: Ensure contributions from all group members.
- **Submission**: Submit your report and supplementary materials via Canvas.

Note: Decide on the most appropriate methods for each part of the analysis. If you need to make additional assumptions, write them down and explain them shortly.